

Clarity Scheduling Business Model and Configurable Rules

Operational rule structure for sales discovery, hospital review, and IT implementation.

Business Model

Business layer	What the company offers	Hospital-facing value	IT build implication
Discovery framework	A structured interview and rule checklist for every program.	Captures hidden scheduling logic that usually lives in chiefs' heads and Excel comments.	Rules engine must be configurable, not hardcoded.
Data model	Residents, institutions, rotations, blocks, services, shifts, requests, attendance, PTO, call switches.	One source of truth replaces scattered forms, PDFs, and spreadsheets.	Relational backend with audit trail and permissions.
Scheduling engine	Builds master schedules and monthly service schedules from rules and approved inputs.	Saves chiefs from rebuilding from scratch and exposes gaps early.	Constraint solver/draft generator plus manual chief editing.
Workflow platform	Approvals, call switches, PTO, attendance, publishing, notifications.	Reduces manual PDF searching and three-person coordination for simple changes.	State machines for requests and schedule publishing.
Transition design	Digital views plus Excel-like views, import existing Excel/PDF, export schedules.	Meets chiefs where they are instead of forcing immediate behavior change.	Reusable view layer and upload/mapping tools.
Analytics layer	Coverage, fairness, hours, attendance, call balance, vacation capacity.	Leadership sees risk and workload distribution before issues become conflicts.	Metrics generated from schedule facts, not manual summaries.

Critical Linkage Model

Source event	Feeds into	Modules affected	Why linkage matters
Resident submits annual ranking	Chief master builder	Resident profile, master schedule, service coverage	Preferences become decision support for the annual source schedule.
Chief approves vacation/request	Master/monthly builder constraints	Resident profile, schedule builder, analytics	Only approved requests should influence generated schedules.
Chief finalizes master schedule	Official source of rotations by block	Service coverage, call pools, resident portal, monthly builder	Monthly schedules must pull real assigned residents, not random samples.
Institution clinic/didactic rule changes	Protected time generator	Resident profiles, monthly builder, call switch checks, attendance	Outside rotator rules apply automatically without entering each resident manually.
Service rule changes	Monthly schedule builder	Coverage targets, shift templates, alerts, analytics	Each service can have different rules, roles, and staffing needs.
Call switch accepted and approved	Published schedule update	Both residents' schedules, service tab, call counts, chief audit trail	Prevents manual PDF review and missed clinic/back-to-back conflicts.
Attendance check-in	Didactics record	Resident profile, chief analytics, attendance dashboard	Lecture attendance updates instantly for both resident and chief view.
PTO/sick leave entered	Availability and coverage	Resident profile, monthly builder, service coverage, analytics	Chiefs see service impact and resident totals.

Hospital Rule Categories

Rule category	#	Priority	Example question
Analytics/Alerts	4	MVP	Which coverage alerts are required?
Call Switch	5	MVP+	Can residents offer a call for switch?
Design/UX	6	MVP, MVP+	Should chief and resident views be separate?
Didactics/Attendance	5	MVP, MVP+	Which didactics require residents to leave service?
Holidays	4	MVP+	Which holidays require special staffing?
Institution Rules	4	MVP, MVP+	How are clinic days assigned?
Master Schedule	12	MVP, MVP+	Which rotations are mandatory for each PGY class?
Monthly Scheduler	9	MVP	When building a monthly schedule, how are residents pulled?
Night/Call/Clinic	8	MVP	Which residents can do nights on each service?
PTO/Leave	4	MVP, MVP+	What PTO fields are tracked?
Program Setup	8	MVP, MVP+	How many scheduling blocks exist per academic year?
Requests	4	MVP	What request types can residents submit?
Resident Intake	10	MVP, MVP+	When do residents submit annual preferences?
Resident Profile	5	MVP	What fixed resident facts must be collected once?
Service Coverage	5	MVP	Should there be a service coverage workbook separate from monthly schedules?

Program Setup

ID	Rule / question	Linked modules	Editable	Acceptance
PGM-001	How many scheduling blocks exist per academic year?	All block selectors, request deadlines, master schedule, monthly builders, analytics	Chief can edit before launch	Changing block count updates every annual and monthly workflow.
PGM-002	What are the exact start and end dates for each block?	Resident request deadlines, service schedules, holidays, PTO, attendance	Chief editable	Every block has date range and no gaps/overlap unless intentionally allowed.
PGM-003	How many days before a block must residents submit requests?	Resident request forms, reminders, chief approvals	Chief editable	Late requests are flagged but chiefs can override.
PGM-004	Which trainee groups need schedules?	Profiles, master schedules, service coverage, eligibility	IT config plus chief edit	Every resident belongs to one class/year/track at a time.
PGM-005	Does the program use tracks with different curriculum requirements?	Master builder, elective eligibility, service minimums	Chief editable	Track-specific requirements appear in master builder and resident profile.
PGM-006	Which institutions send residents or outside rotators?	Protected time, didactics, clinics, eligibility, call rules	Chief editable	Every resident has an institution profile or an exception flag.
PGM-007	Who can view, edit, approve, publish, and export schedules?	All portals, audit log, request decisions	IT config	Residents cannot see private master schedules of others.
PGM-008	Can the department upload an already completed master schedule?	Profiles, master schedule, service coverage, monthly builder	Chief runs import; IT builds parser	Imported assignments populate profiles and builder source data.

Resident Intake

ID	Rule / question	Linked modules	Editable	Acceptance
INT-001	When do residents submit annual preferences?	Chief master builder	Chief editable	Every annual submission has timestamp and status.
INT-002	How many master schedule templates must residents rank?	Master builder rankings step	Resident submits; chief reviews	Duplicate ranking numbers are blocked or flagged.
INT-003	Can two residents request the same top master template?	Master builder conflict view	Automated flag; chief decides	System highlights competing top choices and decision factors.
INT-004	Should first-come-first-serve affect priority?	Master builder sorting and priority notes	Chief override	Chief can sort by submission time and override with documented reason.
INT-005	Do fellowship applicants receive priority for early electives?	Master builder and elective assignment	Chief decides	Fellowship flag appears beside rankings and requests.
INT-006	How are major life events handled?	Requests center, master builder, PTO/LOA	Chief approves	Major life event requests are visible but controlled by privacy policy.
INT-007	Do residents rank holidays they prefer off?	Holiday workbook	Resident submits; chief approves	Holiday request ranks feed holiday staffing and conflict view.
INT-008	How many elective choices are ranked?	Elective assignment and master builder	Resident submits; chief approves	Elective choices have rank, eligibility, availability, and capacity.
INT-009	Can residents request split two-week electives?	Master/elective builder	Chief editable	Split block requests are visible and do not silently break vacation rules.
INT-010	Should free-text comments be structured for chiefs?	Chief review panel	Chief can retag	Comments are searchable and linked to resident profile.

Master Schedule

ID	Rule / question	Linked modules	Editable	Acceptance
MAS-001	Which rotations are mandatory for each PGY class?	Master builder, service coverage, profile	Chief editable	Core rotations can be added, renamed, dragged, deleted, and mapped to services.
MAS-002	Which rotations are elective/vacation/call-pool eligible?	Call pool, vacation capacity, master builder	Chief editable	Elective rotations are marked for vacation/call eligibility separately.
MAS-003	How do requirements differ by PGY?	Master builder and eligibility checks	Chief editable	Each PGY has its own core/elective/minimum rule set.
MAS-004	How many master schedules must be built for each class?	Master builder rankings	Chief editable	System flags mismatch between resident count and template count.
MAS-005	For each block, what minimum and maximum residents are needed per rotation/service?	Block coverage, service coverage, alerts	Chief editable	Counts per block and service show OK/short/over.
MAS-006	How many vacation/elective slots are allowed per block?	Vacation requests, master builder	Chief editable	Vacation requests cannot silently exceed block capacity.
MAS-007	Should electives be distributed to reduce burnout?	Master builder suggestions	Chief editable	System flags residents with long inpatient-heavy stretches.
MAS-008	Should residents ending a block on nights avoid inpatient service next block?	Master builder conflict inspector	Chief override	Unsafe night-to-inpatient transition is flagged.
MAS-009	Can chiefs override generated master assignments?	Master grid, audit trail, profiles	Chief editable	Manual changes update service coverage and resident profiles.
MAS-010	When is master schedule considered official source data?	Service coverage, monthly builders, resident portals	Chief publishes	Downstream builders use only finalized or explicitly selected draft source.
MAS-011	Do chiefs need an Excel-like view of the annual master?	Master schedule UI	User toggle	Digital and Excel-like views show same source data.
MAS-012	How should an existing Excel master be reviewed after upload?	Profiles, service coverage, call pools	Chief approves import	No imported data becomes official until chief approves mapping.

Service Coverage

ID	Rule / question	Linked modules	Editable	Acceptance
SCV-001	Should there be a service coverage workbook separate from monthly schedules?	Monthly service builder, overview, analytics	Automated with chief edits	Service coverage names are pulled from finalized master, not random.
SCV-002	Can users click service counts to see names?	Resident profiles, service detail panel	Automated	Clicking a service shows assigned residents and profile links.
SCV-003	Should counts be visible across all blocks?	Chief overview and service coverage	User toggle	Chief can see block-service staffing across the year without crowding.
SCV-004	How are outside rotators counted?	Service coverage, institution rules, monthly builder	Chief editable	Outside rotator load is tracked by block and service.
SCV-005	When should alerts appear?	Overview, master builder, service coverage	Automated	Alerts show service, block, assigned count, target, and names.

Monthly Scheduler

ID	Rule / question	Linked modules	Editable	Acceptance
MON-001	When building a monthly schedule, how are residents pulled?	Monthly service schedule	Automated with chief override	PICU Block 2 pulls only residents assigned to PICU Block 2 plus eligible supplements.
MON-002	Does each service have its own rules?	Monthly generator and validation	Chief editable	Each service tab can be configured independently.
MON-003	Can services have units/teams/locations?	Schedule builder and analytics	Chief editable	Locations can be added, renamed, colored, and given minimums.
MON-004	Can services define custom roles?	Shift cards, schedule builder, fairness stats	Chief editable	Roles have time, color, eligibility, count target, and fairness category.
MON-005	What timing fields are required for a shift?	Monthly builder, hours analytics	Chief editable	Hours calculate from shift templates and manual overrides.
MON-006	Should clinics/didactics/exams auto-place before shifts?	Monthly builder conflict protection	Automated; chief can edit	Protected time appears before assignments and blocks unsafe calls.
MON-007	What fairness metrics are balanced?	Insights and analytics	Chief config	Chief sees imbalance before publishing.
MON-008	Can chiefs drag/drop and override generated schedules?	Monthly schedule, audit trail, resident view after publish	Chief editable	Manual edit recalculates coverage and conflict warnings.
MON-009	When do residents see monthly schedules?	Resident portal, published schedules	Chief publishes	Drafts are hidden from residents unless shared intentionally.

Night/Call/Clinic

ID	Rule / question	Linked modules	Editable	Acceptance
NCL-001	Which residents can do nights on each service?	Monthly builder	Chief editable	Ineligible residents are blocked or warned.
NCL-002	What is the maximum consecutive night stretch?	Monthly builder, fairness stats	Chief editable	Generator avoids exceeding max; chief override flagged.
NCL-003	What post-call/rest rules apply?	Monthly builder, resident schedule	Chief editable	Post-call rest appears automatically when rule requires.
NCL-004	Can residents be assigned call on clinic day?	Call switch and monthly builder	Chief editable	Call on clinic day is flagged or blocked.
NCL-005	Can residents take call the day before clinic?	Call switch and monthly builder	Chief editable	Call before clinic is flagged or blocked.
NCL-006	Can residents have back-to-back calls?	Call switch and monthly builder	Chief override	Back-to-back call request shows reason and warning.
NCL-007	How is golden weekend defined?	Fairness stats and resident portal	Chief editable	System counts golden weekends per resident per block/year.
NCL-008	What weekend targets apply?	Monthly builder analytics	Chief editable	Fairness panel compares residents on same service/block.

Call Switch

ID	Rule / question	Linked modules	Editable	Acceptance
CSW-001	Can residents offer a call for switch?	Call switch marketplace, chief approval	Resident submits	Offered call appears to eligible residents.
CSW-002	What checks run before another resident accepts?	Resident switch view and chief approval	Automated	Every switch shows eligibility reasons before acceptance.
CSW-003	Can eligible residents accept offered calls?	Chief request center	Resident accepts; chief approves	Switch status moves offered -> accepted -> chief review -> approved/declined.
CSW-004	What should chiefs see?	Chief approvals and schedules	Chief final decision	Chief can approve, decline, or request changes.
CSW-005	What updates after switch approval?	Resident portal, monthly schedule, analytics	Automated	No manual PDF searching is needed after approval.

Requests

ID	Rule / question	Linked modules	Editable	Acceptance
REQ-001	What request types can residents submit?	Resident profile, chief approvals, schedule builders	Resident submits; chief approves	Each request has type, date/block, reason, priority, status.
REQ-002	What request states exist?	All request screens and builder inputs	Chief action	Only approved/applied requests become generator constraints.
REQ-003	How are competing requests handled?	Approval center and master builder	Chief decides	System suggests priority but does not force final decision.
REQ-004	Should pending requests appear during schedule building?	Master and monthly builders	Automated	Chief can see whether request is approved/pending/declined while building.

PTO/Leave

ID	Rule / question	Linked modules	Editable	Acceptance
PTO-001	What PTO fields are tracked?	Resident profile, monthly builder, analytics	Chief approves	Resident and chief see running totals and affected services.
PTO-002	How are sick days recorded?	Profile, service schedule, analytics	Chief finalizes	Sick leave updates profile and coverage alerts.
PTO-003	How are longer leaves handled?	Master, service coverage, monthly builders	Chief restricted edit	LOA appears as protected/unavailable without exposing private details unnecessarily.
PTO-004	How are one-time protected events entered?	Monthly builder and profile	Chief editable	Events block unsafe assignments and show in resident schedule.

Didactics/Attendance

ID	Rule / question	Linked modules	Editable	Acceptance
DID-001	Which didactics require residents to leave service?	Monthly builder and attendance	Chief editable	Morning reports that do not affect coverage can be optional/untracked.
DID-002	Do institutions have recurring didactic patterns?	Resident profile, monthly builder, attendance	Chief editable	Outside rotator didactics auto-apply based on institution.
DID-003	How is attendance captured?	Resident/chief profiles and analytics	Chief creates session	Checking a resident updates attendance count automatically.
DID-004	What fields define a lecture?	Attendance analytics	Chief editable	Session can be reused or copied.
DID-005	What attendance stats are shown?	Resident profile, chief analytics	Automated	Chief can see missed count and context for each resident.

Holidays

ID	Rule / question	Linked modules	Editable	Acceptance
HOL-001	Which holidays require special staffing?	Holiday workbook and resident requests	Chief editable	Holiday list can be customized by institution.
HOL-002	Do residents rank preferred holidays off?	Holiday workbook	Resident submits; chief approves	Holiday requests appear beside staffing needs.
HOL-003	Which services need holiday coverage?	Holiday workbook	Chief editable	Each holiday has service-specific minimum counts.
HOL-004	Does program have one- or two-week holiday break schedule?	Holiday breaks workbook	Chief editable	Holiday break schedule links to master and service coverage.

Institution Rules

ID	Rule / question	Linked modules	Editable	Acceptance
INS-001	How are clinic days assigned?	Monthly builder and call switch	Chief editable	Clinic patterns protect time and block unsafe calls.
INS-002	How are didactics assigned?	Monthly builder and attendance	Chief editable	Protected didactics appear automatically in schedule builder.
INS-003	Which services/roles can outside rotators cover?	Service coverage and monthly builder	Chief editable	Ineligible outside rotators are not auto-assigned.
INS-004	Do some institutions have special PICU/NICU didactic or off-service rules?	PICU/NICU schedule builder	Chief editable	Institution-specific exceptions appear when that resident is pulled.

Analytics/Alerts

ID	Rule / question	Linked modules	Editable	Acceptance
ANL-001	Which coverage alerts are required?	Overview, builders, service coverage	Automated	Alerts are clickable and explain cause/action.
ANL-002	Which fairness summaries are required?	Resident profile, chief analytics	Automated	Resident totals can be compared to peer group.
ANL-003	How should work hours be calculated?	Resident/chief analytics	Automated	Hours recalculate when chiefs edit shifts.
ANL-004	What readiness status should each schedule show?	Overview and schedules tab	Automated with chief control	Overview shows each block and service status.

Design/UX

ID	Rule / question	Linked modules	Editable	Acceptance
DES-001	Should chief and resident views be separate?	All UI	IT config	Chief and resident portals have clear toggle/roles.
DES-002	How do we avoid crowded screens?	Master, service coverage, call pools, holidays	Design standard	Large Excel-like data can be explored without overwhelming default view.
DES-003	Where should Excel-like views be offered?	View toggles	User toggle	Digital and Excel-like views reflect the same data.
DES-004	Should resident names/initials link to profiles?	All roster/count views	Design standard	Back navigation preserves where user came from.
DES-005	Can departments customize colors/labels?	Schedule UI and exports	Chief editable	Design customization does not break analytics categories.
DES-006	How should uploaded Excel/PDF data be reviewed?	Master/service/call/holiday modules	Chief approves	No upload silently overwrites official data.

Use the editable workbook for live hospital discovery

The PDF is a readable playbook. The Excel workbook is the actual review/edit tool with fields for hospital answer, IT notes, and implementation status.